

INDIAN STATISTICAL INSTITUTE

Parametric Inference

(B.Stat. 3rd year)

Semester 1, 2026–27

Homework 1

1. Let X have p.m.f.

$$f(x | \theta) = \frac{\Gamma(\alpha + x)}{\Gamma(\alpha) \Gamma(x + 1)} \theta^\alpha (1 - \theta)^x, \quad (x = 0, 1, 2, \dots)$$

where $0 < \theta < 1$ is an unknown parameter, $0 < \alpha$ is known and $\Gamma(\cdot)$ is the Gamma function. For any given β , does there exist UMVUE of θ^β ? Find the UMVUE when it exists.

2. Let X be Poisson with unknown mean $\theta > 0$. Find the UMVUE of $e^{\beta\theta}$ for any given $-\infty < \beta < \infty$.
3. Let $X_1, X_2, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} N(\theta, 1)$, $-\infty < \theta < \infty$ unknown parameter. Find the UMVUE of $\Pr\{X_1 > 0\}$.
4. For a given sequence $\{a_n\}$ of positive constants, let

$$f(\theta) = \sum_{x=0}^{\infty} a_x \theta^x < \infty$$

for $0 < \theta < b$. θ is some unknown parameter. Let X have p.m.f. $\frac{a_x \theta^x}{f(\theta)}$. Does the UMVUE of $E_\theta(X)$ attain Cramér–Rao Lower Bound? Verify.

5. Let $Y_1, Y_2, \dots, Y_n$ be independent observations such that

$$Y_i = N(\beta x_i, \sigma^2), \quad i = 1, 2, \dots, n$$

where $-\infty < \beta < \infty$ and $0 < \sigma^2 < \infty$ are unknown parameters and $x_1, x_2, \dots, x_n$ are known non-zero constants. Show that the least squares estimate of β and the associated residual variance are UMVUE's for β and σ^2 respectively. Are the two estimates independent?

6. $X_1, \dots, X_n \stackrel{\text{i.i.d.}}{\sim} f(x | \theta) = \frac{1}{\theta} e^{-x/\theta}$, $0 < x, \theta < \infty$. Show that $\sum_{i=1}^n X_i$ and $\sum_{j=1}^p X_j / \sum_{k=1}^q X_k$ are independent for $1 \leq p \neq q \leq n$.